CALCULUS · DERIVATIVES

Secant to Tangent

Explore & Reflect TEACHER KEY

Name _____ Date _____ Period _____

Open **Secant to Tangent** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. For $f(x) = x^2$, what is the slope of the secant line from $x = 1$ to $x = 3$?
   $\frac{(9 - 1)}{2} = 4$
2. As the second point moves toward $x = 1$, what do you think happens to the secant slopes?
   They approach 2, the slope of the tangent line at x = 1.

2 Explore use the tool

1. $f(x) = x^2$, $a = 1$. Record the secant slope for $h = 1$, 0.1 and 0.01.
   3, 2.1, 2.01
2. Now use $h = -0.1$ and $h = -0.01$. What do the slopes approach from the left?
   1.9 and 1.99; they approach 2.
3. For $|x|$ at $a = 0$, record the slopes for $h = 0.01$ and $h = -0.01$. Why is there no derivative there?
   1 and –1. The one-sided limits differ (a corner), so there is no single tangent slope.
4. For sin x, visit at least 8 points and build the slope graph. Which function does it look like?
   cos x

3 Record use the tool and fill in the table

Secant slopes from both sides

Round to 4 decimal places.

f(x)	a	slope with h = 0.01	slope with h = −0.01	f′(a)
x^2	2	4.01	3.99	4
$x^3 - 3x$	1	0.0301	−0.0299	0
e^x	0	1.0050	0.9950	1
sin x	0	1.0000	1.0000	1

4 Explain what did you discover?

1. The slope of the tangent line at a is the limit of the secant slopes as h → 0.
2. h can't equal 0, because a secant line needs two different points.
3. f is not differentiable at a corner, because the left and right secant slopes approach different numbers.
4. Where f has a horizontal tangent line, f′(a) = 0.

5. Explain how the graph of f′ for $f(x) = x^2$ is built from tangent slopes, and why it is a straight line.
   At each point a the tangent slope is 2a. The points (a, slope) are (–1, –2), (0, 0), (1, 2), (2, 4), …, which lie on the line y = 2x.

5 Apply a new situation, no tool

1. Use the limit definition to find f′(3) for $f(x) = x^2 - x$.
   $\lim_{h \to 0} \frac{((3+h)^2 - (3+h) - 6)}{h} = \lim_{h \to 0} \frac{(5h + h^2)}{h} = 5$
2. f(1.9) = 6.62, f(2) = 7, f(2.1) = 7.42. Estimate f′(2).
   $\frac{(7.42 - 6.62)}{0.2} = 4$ **(one-sided estimates: 4.2 and 3.8)**
3. Where is g(x) = |x – 3| not differentiable? Why?
   At x = 3: a corner, with left slope –1 and right slope 1.
4. For $f(x) = x^3 - 3x$, where is the tangent line horizontal?
   x = –1 and x = 1

Teacher notes

Intended discovery: The derivative is the limit of secant slopes as h → 0 from both sides; when the one-sided limits differ there is no derivative, and the tangent slopes at every point form the graph of f′.

Likely misconceptions:

- Setting h = 0 directly ($\frac{0}{0}$).
- Checking only h > 0.
- Continuous means differentiable (|x| is continuous but not differentiable at 0).

Answers that can vary: The points students visit when building the slope graph; table values are rounded approximations.